



INTERNAL EXAMINATION : BCA (WINTER)

Subject : 402- IoT (Internet of Things)

Academic Year: 2025-26

Semester: IV

Time : Seat No :

Q.1 (A) Multiple Choice Questions (Any five out of seven)

[5 Marks]

1. Which layer in IoT architecture contains sensors and actuators?
a) Network Layer b) Data Processing Layer
c) Application Layer d) Sensing Layer
2. IPv6 uses how many bits for an IP address?
a) 32 bits b) 64 bits
c) 96 bits d) 128 bits
3. An embedded system typically consists of:
a) Only hardware b) Only software
c) Hardware, software, and RTOS d) Only RTOS
4. Which of the following is NOT a basic building block of an IoT device?
a) Sensing b) Actuation
c) Communication d) Manufacturing
5. What does REST stand for?
a) Remote State Transfer b) Representational State Transfer
c) Real-time State Transmission d) Resource State Transfer
6. In M2M architecture, how many domains are present?
a) Two b) Three
c) Four d) Five
7. Which type of sensor converts physical parameters directly into electrical signals?
a) Digital sensors only b) Smart sensors only
c) All sensors d) Analog sensors only

(B) Short Answer Questions (Any four)

[8 Marks]

1. What is IoT? List any three characteristics of IoT systems.
2. Difference between IPV4 and IPV6.
3. Explain the difference between Analog Sensor and Digital Sensor.
4. What is Big Data.
5. What is Arduino.

Q.2 (A) Long Answer Question

Explain IoT communication models in detail.

[8 Marks]

OR

Describe the Raspberry pi in detail.

Q.2 (B) Long Answer Question

[5 Marks]

Explain the four-layer IoT architecture.

Q.3 Descriptive Question. (Any Two)

[12 Marks]

(A) Describe the Big Data Analytics in detail.

(B) Explain different types of actuators used in IoT systems.

(C) Describe the components of Wireless Sensor Networks (WSN).

Q.4 Write short notes (Any Two)

[12 Marks]

(A) Difference Between IoT & M2M.

(B) What is an IoT Device? Explain basic building blocks of an IoT Device.

(C) Case Study - Smart City Applications.